

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. What two roles do mastication and taste play in starting digestion, and which cranial nerves carry taste from the anterior 2/3 vs posterior 1/3 of the tongue?
2. Which salivary gland provides ~70% of resting saliva volume — is it the parotid or the submandibular?
3. Is salivary secretion controlled by nerves, hormones, or both?

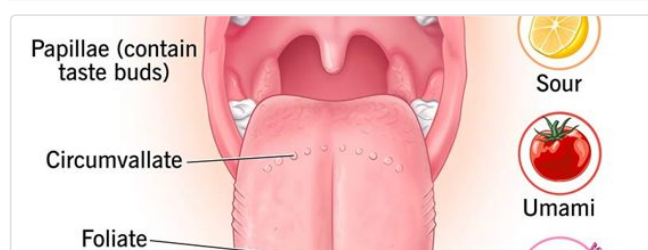
Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Functions of the Oral Cavity

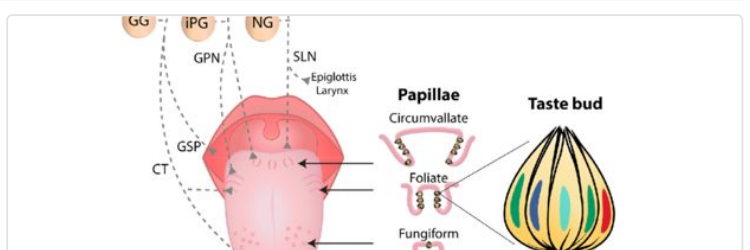
Function	What it does	Key structures / pathway
Mastication	↑ food surface area for enzymatic digestion	Periodontal pressure receptors → brainstem chewing centre → motor nuclei of CN V, VII, IX, XII → masseter, temporalis, pterygoids
Taste (gustation)	Sensory analysis; drives the salivary reflex	Taste buds (fungiform, circumvallate, foliate) — sweet/sour/salty/bitter/umami
Lubrication & bolus formation	Moistens food; mucins coat particles for safe swallowing	Saliva mucins
Defense	Immune surveillance against pathogens entering with food	Tonsillar tissue; salivary IgA, lysozyme, lactoferrin
Speech	Airflow & resonance shaping for articulation	Tongue and lips

2 Two reflex chains worth knowing cold

Mastication reflex	Periodontal pressure receptors → brainstem chewing centre → motor nuclei of CN V, VII, IX, XII → masseter / temporalis / medial & lateral pterygoids contract
Taste → salivation	Taste bud stimulated → CN VII (anterior 2/3 tongue) / CN IX (posterior 1/3) / CN X (epiglottis, pharynx) → nucleus tractus solitarius → hypothalamus & salivatory nuclei → salivation stimulated



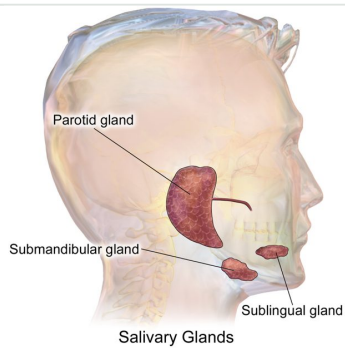
Taste bud distribution and the five modalities across tongue regions.



Gustatory pathway converging on the rostral nucleus tractus solitarius (rNTS).

3 Salivary Glands — anatomy

Gland	Secretion type	Contribution
Parotid	Serous, watery	Rich in amylase
Submandibular	Mixed	~70% of resting saliva volume — the main contributor, not the parotid
Sublingual	Mucous-rich	Small volume, thick secretion
Minor glands	Mixed	Dispersed in lips, cheeks, palate, tongue



The three major salivary glands and their locations.

4 Composition of Saliva

Component	Detail
Water	~99% of saliva
Electrolytes	Na^+ , K^+ , Cl^- , HCO_3^- , Ca^{2+} , phosphate
Enzymes	α -Amylase (starch digestion) · Lingual lipase (fat digestion, active in stomach) · Kallikrein (forms bradykinin → ↑ blood flow)
Mucins	Lubrication
Antimicrobial	Lysozyme, lactoferrin, peroxidase, IgA
Excretory	Urea, uric acid, some drugs

5 Saliva Formation — from acinus to duct

Primary secretion	Acinus produces an isotonic, plasma-like fluid containing amylase and mucin
Ductal modification	Na^+ and Cl^- actively reabsorbed ; K^+ and HCO_3^- secreted ; ducts are impermeable to water
Result	Final saliva is always hypotonic to plasma — regardless of flow rate

Flow rate	Composition
Low flow	More hypotonic, acidic, ↑ K^+
High flow	Closer to plasma composition, alkaline, ↑ Na^+ & HCO_3^-

6 Control of Salivary Secretion

NEVER MISS: salivation is under **neural control only** — unlike most GI secretions, there is no hormonal regulation of the salivary glands.

Division	Pathway	Transmitter / receptor	Effect
Parasympathetic	CN VII (chorda tympani → submandibular ganglion) · CN IX (otic ganglion → parotid)	ACh → muscarinic M3	Copious, watery saliva
Sympathetic	T1-T3 → superior cervical ganglion	NE → β -adrenergic	Smaller volume, more mucous & protein-rich

7 Functions of Saliva — the summary line

Function	Detail
Digestion	Begins carbohydrate & lipid digestion
Lubrication	For swallowing & speech
Protection	Antimicrobial, buffer, remineralization of teeth

Taste	Dissolves substances for chemoreceptors
Excretion	Minor route for some waste

P Must-remember pearls

- The **submandibular** gland, not the parotid, supplies ~70% of resting saliva volume.
- Salivation is under **neural control only** — no hormone regulates it.
- Parasympathetic (M3) → copious watery saliva; sympathetic (β -adrenergic) → small volume, protein-rich saliva.
- Ducts are impermeable to water, so final saliva is **always hypotonic** to plasma, at any flow rate.
- Taste: anterior 2/3 tongue = CN VII; posterior 1/3 = CN IX; epiglottis/pharynx = CN X.

Q Test yourself

1. A patient with an isolated chorda tympani nerve injury would most likely show:

- | | |
|---|---|
| A. Loss of taste from the posterior 1/3 of the tongue | B. Reduced parasympathetic (watery) salivation from the submandibular gland |
| C. Loss of all parotid secretion | D. Loss of the mastication reflex |

2. A muscarinic antagonist (e.g. atropine) given to a patient would be expected to cause:

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|---|--|
| A. Increased watery saliva | B. Dry mouth from blocked M3 receptors |
| C. No change, since saliva is hormonally controlled | D. Increased mucous secretion only |

3. Which gland contributes the largest share of saliva at rest, between meals?

- | | |
|---------------|------------------|
| A. Parotid | B. Submandibular |
| C. Sublingual | D. Minor glands |

4. During rapid chewing (high salivary flow), the saliva produced is:

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|------------------------------|--|
| A. More hypotonic and acidic | B. Closer to plasma composition and alkaline |
| C. Isotonic with no ions | D. Free of bicarbonate |

5. A lesion of CN IX would impair taste sensation in which region?

- | | |
|-------------------------------|--|
| A. Anterior 2/3 of the tongue | B. Posterior 1/3 of the tongue |
| C. The epiglottis only | D. No taste loss — CN IX is purely motor |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — B** — the chorda tympani carries parasympathetic fibres to the submandibular ganglion (and taste from the anterior 2/3 tongue, but the salivary effect is the testable point here).
- 2 — B** — M3 blockade removes the parasympathetic drive that produces watery saliva.
- 3 — B** — the submandibular gland supplies ~70% of resting saliva.
- 4 — B** — high flow saliva approaches plasma composition and is alkaline ($\uparrow \text{Na}^+$, $\uparrow \text{HCO}_3^-$).
- 5 — B** — CN IX (glossopharyngeal) carries taste from the posterior 1/3 of the tongue.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam